Adding Integers: Introduction

You can attempt each problem as many times as needed. To check your answers, click on the blue box with a question mark, or use the Enter key on your keyboard. If you get an answer wrong, you will see a red x and you can try again. ***Do not inlcude commas in your answers.***

The Set of Integers

In this lesson, you will learn how to add numbers that belong to the set of integers. The set of **integers** consists of the whole numbers along with the negative counting numbers.

Counting Numbers: $\{1, 2, 3, \dots\}$

Whole Numbers: $\{0, 1, 2, 3, \dots\}$

Integers: $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Extra Insight: False Numbers

Although negative numbers were understood in China more than 2000 years ago and later in India, they were not immediately accepted in Europe. For many centuries, European mathematicians were uncomfortable with the idea of numbers less than zero and sometimes referred to them as false or absurd numbers. However, during the 16th and 17th centuries, as algebra and calculus developed, the importance of negative numbers became undeniable. Today, negative numbers are an essential part of mathematics and are commonly used to represent quantities such as losses, debts, temperatures below zero and elevations below sea level.

Adding Integers

To understand how to add integers, we will begin with the Chinese Red-Black Rule and then a number line to visualize addition. Then we will use the concepts of assets and debts to understand addition in a real-world context. These approaches will lead us to two rules: one for adding integers with the different signs and another for adding integers with the same signs.

Author(s): Kelly Stady

The Red-Black Rule

Negative numbers were first used systematically in ancient China. In *The Nine Chapters on the Mathematical Art* (written between about 200 BC and 100 AD), Chinese mathematicians used counting rods to represent numbers. Red rods were used for positive numbers and black rods for negative numbers. For example, the number 3 was represented by three red rods and the number -2 was represented by two black rods. These rods were used as a calculator of sorts for financial transactions involving profits and losses and to solve simple equations. The Red-Black Rule is described below.

(a) A red rod represents 1 and a black rod -1.

(b) To find the total of profits (red) and losses (black), begin by placing rods on a board to represent each number.

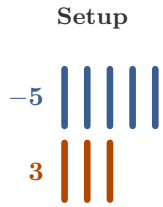
(c) Since a profit and an equal loss result in 0, gather all pairs of red and black rods and remove them.

(d) The total is what remains.

Since red and black can be difficult for some people to distinguish, we will use **orange rods** to represent **positive numbers (+)** and **blue rods** to represent **negative numbers (–)**. We will now refer to this rule as the Orange-Blue Rule.

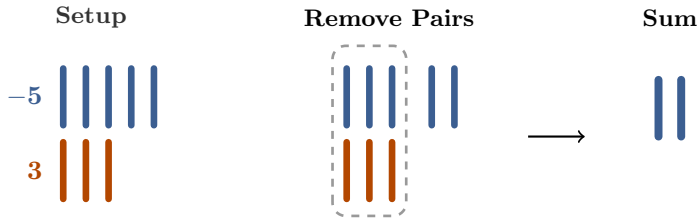
Example 1. Use the Orange-Blue Rule to find the sum $-5 + 3$.

Explanation. We begin with **five blue rods** to represent the number -5 and **three orange rods** to represent the number 3.



Alt text: The "Setup" section showing five blue vertical bars representing negative 5 and below them are three orange vertical bars representing 3.

Next, we remove all pairs of orange and blue rods. The sum is what remains.



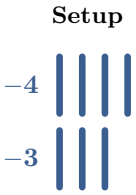
Alt text: A diagram modeling the addition of a negative number and a positve number using vertical color-coded rods. In the "Setup" section, on the left, five blue bars represent negative 5 and below are three orange bars representing 3. In the "Remove Pairs" section, in the middle, the three orange bars are paired vertically with three of the blue bars inside a dashed box to show they cancel each other out, leaving two blue bars outside the box. A black arrow points to "Sum" section which shows the two remaining blue bars.

Count the number of bars that remain to find the sum. Remember to use the correct sign.

$-5 + 3 = -2$

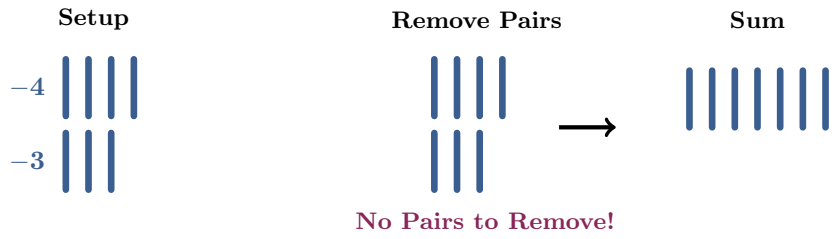
Example 2. Use the Orange-Blue Rule to find the sum $-4 + (-3)$.

Explanation. We begin with **four blue rods** to represent the number -4 and **three blue rods** to represent the number -3.



Alt text: The "Setup" section showing four blue vertical bars representing negative 4 and below them are three blue vertical bars representing negative 3.

Since we started with only blue rods, there are no orange-blue pairs to remove.



Alt text: A diagram modeling the addition of two negative numbers using vertical color-coded rods. In the "Setup" section, on the left, there are four blue bars representing negative 4 and below are three blue bars representing negative 3. The "Remove Pairs" section, in the middle, is the same as the "Setup" section. A caption below the rods reads "No Pairs to Remove!" because all rods are the same color. A black arrow points to the "Sum" section, which shows all seven blue rods together.

Count the number of bars that remain to find the sum. Remember to use the correct sign.

$-4 + (-3) = -7$

Problem 1 *Use the Orange-Blue Rule to find the sum $-2 + 6$. The problem has been started for you below.*



Alt text: A diagram modeling the addition of a negative number and a positive number using vertical color-coded rods. In the "Setup" section, on the left, there are two blue vertical bars representing negative 2 and six orange rods representing 6. The "Remove Pairs" section, on the right, is the same as the "Setup" section.

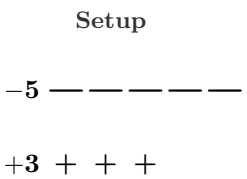
After you remove the pairs, count the number of bars that remain to find the sum. Remember to use the correct sign.

$-2 + 6 = 4$

The use of colors in the Orange-Blue Rule is not necessary. Below, the first example has been redone using a plus sign (+) for positive 1 and a minus sign (−) for −1. You can also use things like different colored poker chips or pennies and dimes. Try modifying the rule, using your own way to represent positive and negative numbers. Be creative!

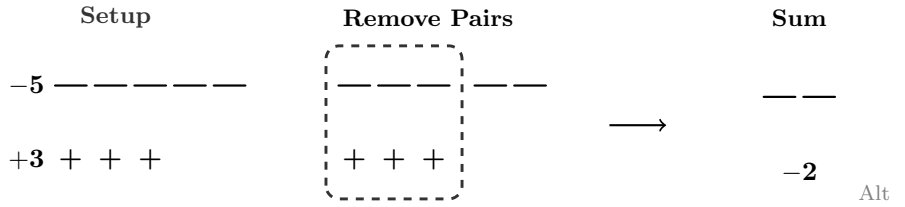
Example 3. Find the sum $-5 + 3$ using a plus sign (+) for 1 and a minus sign (−) for −1.

Explanation. We begin with **five minus signs** (−) to represent the number −5 and **three plus signs** (+) to represent the number 3.



Alt text: The 'Setup' showing five minus signs to represent −5 and below three plus signs represent 3.

Next, we remove all plus-minus pairs. The sum is what remains.



Alt text: A diagram modeling the addition of a negative number and a positive number using plus and minus signs. In the "Setup" section, on the left, five minus signs represent negative 5 and below them three plus signs represent 3. In the "Remove Pairs" section, in the middle, three minus signs and three plus signs are grouped inside a dashed box to show that they cancel. Two minus signs remain outside the box. A black arrow points to the "Sum" section which shows two minus signs labeled negative 2.

$-5 + 3 = -2$

Although this modified rule isn't overly creative, it makes the answer quite clear.